

CS & IT ENGINEERING



Computer Network - 2

Transport & Application Layer

Lecture No. - 03

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Recap of Previous Lecture



Topic

Socket Programming





Topics to be Covered



Topic

TCP Options

Topic

TCP Window Scale

Topic

TCP Timers



Topic : Socket Programming with UDP

Unconnected
UDP Socket;



UDP Client

socket()

→ sendto()

← recvfrom()

close()

UDP Server

socket()

bind()

→ recvfrom()

sendto()

close()

UDP segment

UDP segment

} loop

[MCQ] [GATE-2017, Set-2, 2-Marks] $\Rightarrow \#$



#Q. Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT?

- ☒ I. A connected UDP socket can be used to communicate with multiple peers simultaneously. FALSE
- ☒ II. A process can successfully call connect function again for an already connected UDP socket. TRUE

- ☐ A I only
- ☒ B II only
- ☐ C Both I and II
- ☐ D Neither I nor II

Ans: B



Topic : Connected UDP sockets



UDP Client

socket()

connect()

write()

read()

close()

UDP Server

socket()

bind()

read()

write()

close()

UDP Segment
Payload: Connect Message

UDP Segment

24x7 On



Topic : TCP Optional Header

→ Also known as "Options" (TCP options) [like IPv4 options]

→ Options are variable in size

→ Maximum size of TCP optional header is 10 words
[Words of 32 bits, i.e. 40 bytes]

→ if options size are not in words
than it uses padding [End-of-Option Option]





Topic : TCP Optional Header

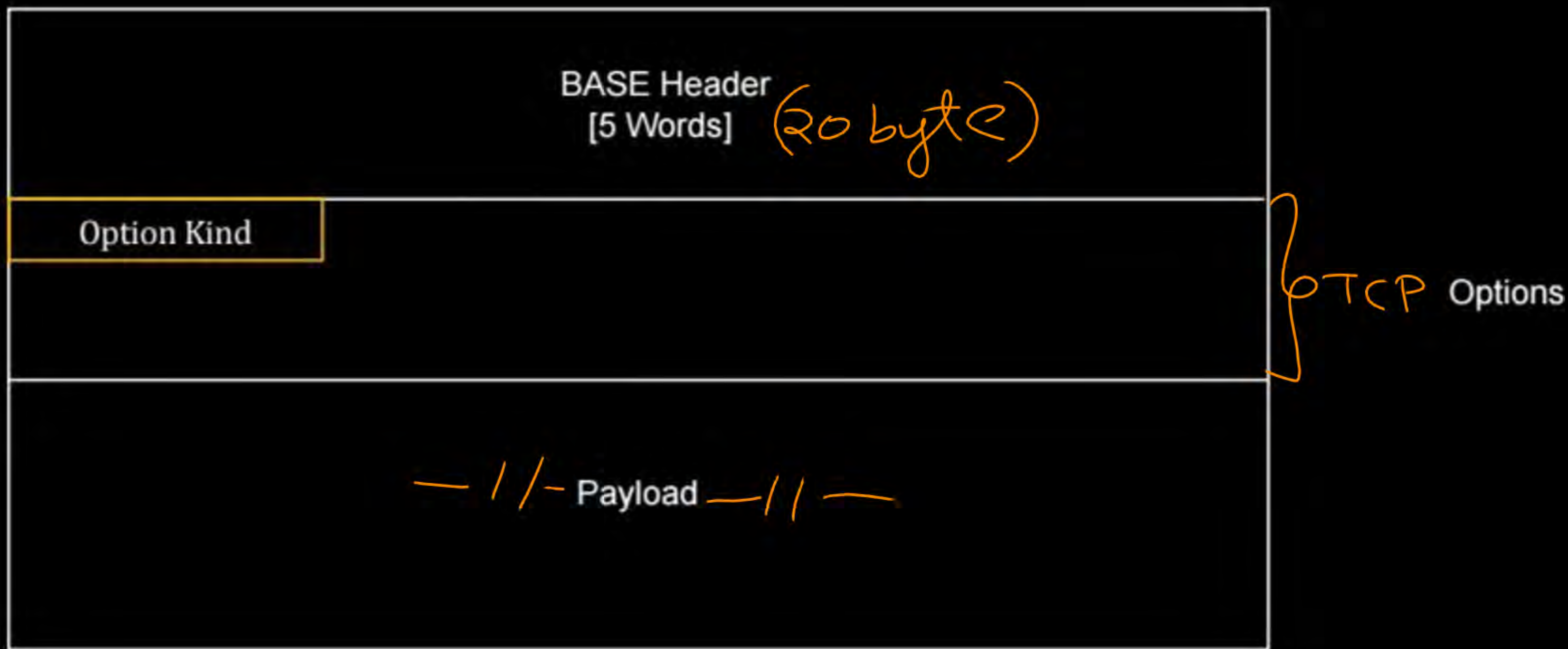


→ Options field component :-

1. Option Kind (1 byte) $\Rightarrow$ (Like IPv4 option type)
2. Option Length (1 byte) $\Rightarrow$ [More than equal to two]
3. Option Data



Topic : TCP Optional Header





Topic : TCP Options



Kind	Length	Purpose
0	—	[End of Option [Padding]]
1	—	No Operation
2	4 bytes	[Maximum Segment Size]
→ 3	3 bytes	Window Scale
4	2 bytes	Selective ACK Permitted
5	10 / 18 / 26 / 34	Selective ACKs (SACK)
8	10	[Timestamp]

} During
connection
establish-
ment



Topic : TCP Optional Header

During Connection Establishment



BASE Header [5 Words]			
SYN = 1			
Option Kind = 2	Option Length = 4	[MSS = Client/Server MSS]	
Option Kind = <u>3</u>	Option Length = <u>3</u>	Shift.cnt = <u>Value</u>	<u>Option Kind = 0</u>

TCP Options



Topic : Bandwidth Delay Product

* Measured RTT



$$\rightarrow \text{bandwidth delay product} = \text{Bandwidth} * \text{RTT}$$

[maximum amount of data that can be in transit on a network]

No. of outstanding bytes
[transmitted, ACK yet to come]

$\Rightarrow$ In TCP, End-to-End Bandwidth is dynamic.

$$\text{RTT} = \text{cycle time} \approx (2t_p)$$

$(t_{p1} + t_{p2})$
 $\rightarrow$ Include all routers
store-and-forward
Delay

$\rightarrow$ For maximum utilization :-

$$\text{Transmit Window Size} = \text{bandwidth delay product}$$

Ex:- $RTT(\text{cycle time}) = 10 \text{ sec}$



$TO \text{ value} > RTT$

End-to-End Bandwidth = 200 bytes/sec

Bandwidth Delay Product = ? (in bytes)

$$= \text{Bandwidth} * RTT$$

$$= 200 \text{ bytes/sec} * 10 \text{ sec}$$

$$= 2000 \text{ bytes}$$

[MCQ] [ISRO-2025, 1-Mark]

#Q. What is the ideal TCP window size of a sender to fully utilize a link of bandwidth 8192000 bps and latency of 32 ms?

☐ A 65536 bytes

☐ B 8192 bytes

☐ C 16384 bytes

☒ D 32768 bytes

$$\text{Bandwidth} = 8192000 \text{ bps} = 2^{13} \times 10^3 \text{ bps}$$

$$\begin{aligned} \text{latency (RTT, cycle time)} &= 32 \text{ ms} \\ &= 2^5 \times 10^{-3} \text{ sec} \end{aligned}$$

$$\begin{aligned} \text{Bandwidth Delay Product} &= \text{RTT} \times \text{Bandwidth} \\ &= 2^5 \times 10^{-3} \text{ sec} \times 2^{13} \times 10^3 \text{ bits/sec} \\ &= 2^{18} \text{ bits} = 2^{15} \text{ bytes} \end{aligned}$$

Ans: D

$$\text{Max}^m \text{ Transmit Window} = \min[\text{CWND}, \text{RWND}]$$



$$\text{Transmit Window Size} \leq \min[\text{CWND}, \text{RWND}]$$

Flow &
Congestion
Control

Congestion
Control

Flow
Control



Topic : Window Size



→ Receiver window size field is 16-bits long
[An advertisement from receiver]

→ Range 0 to $(2^{16}-1)$

→ Define how much data the receiving device can accept more
[Define current available space in receiver's receiving window]

← Recv Buffer

→ Maximum possible available space in receiver's receiving buffer
can be advertised without window scale is $(2^{16}-1)$ bytes

↓
max^m RWND can be $(2^{16}-1)$ bytes



Topic : Window Scaling



→ if bandwidth delay product is greater than $(2^{16} - 1)$ bytes
than window scaling is required to increase throughput

⇒ For high bandwidth networks

Example :-

[NAT]



#Q. Consider a TCP connection where end-to-end approximate bandwidth is 1 Gbps. What can be a maximum value of RTT (in micro-seconds, rounded off to two decimal places) for which no window scaling is required?

Max^m RTT (for which, no window scaling required)

$$= \frac{\text{Max}^m \text{RWND without window scale}}{\text{End-to-End Bandwidth}}$$

$$= \frac{(2^{16} - 1) \text{ bytes}}{1 \text{ Gbps}} = \frac{(2^{16} - 1) * 2^3 \text{ bits}}{10^9 \text{ bits/sec}} = 524.28 \mu\text{s}$$

Ans = 524.28



Topic : Window Scaling



→ Both TCP client and server can advertise their shift_count (shift.cnt)
[during TCP handshake via SYN packet]

→ scale_factor = $2^{\text{shift_count}}$

→ Available space in receiver's receiving buffer (RWND)
= [Advertised Window Size * scale_factor] bytes



Topic : TCP Options for Window Scaling

→ Options field component :-

1. Option Kind (1 byte) = 3
 2. Option Length (1 byte) = 3
 3. Option Data (1 byte) = shift.cnt
- 3 byte ←

Example :-

[NAT]



#Q. Consider a TCP connection with window scaling option and advertised shift count value during TCP connection establishment phase is 2. What can be a maximum possible window size (in bytes) that can be advertised in given scenario?

$$\text{Shift.cnt} = 2$$

$$\text{scale_factor} = 2^{\text{shift.cnt}} = 2^2 = 4$$

$$\begin{aligned} \text{Max}^m \text{ possible RWND} &= [(2^{16} - 1) * \text{scale_factor}] \text{ bytes} \\ &= 262140 \text{ bytes} \end{aligned}$$

(with window scale in given scenario)

$$\boxed{\text{Ans} = 262140}$$



Topic : Window Scaling

$$1 \leq \text{shift_cnt} \leq 14$$



→ Maximum value for shift count can be 14 $\Rightarrow$ TCP

→ Maximum possible available space in receiver's receiving buffer can be advertised with window scale is $(2^{16} - 1) * 2^{14}$ bytes

→ Approx 1GB

In TCP Cong. Control Algo.

$$SSThresh = \infty = (2^{16} - 1) * 2^{14} = \text{Max}^m \text{RWND with window scale}$$

$$CWND = 1MSS$$

[MCQ] [GATE-2015, 2-Mark] IIT-K, H.W. [only one PYQ on BDP]



#Q. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are.

- A 63 milliseconds, 65535×2^{14}
- B 63 milliseconds, 65535×2^{16}
- C 500 milliseconds, 65535×2^{14}
- D 500 milliseconds, 65535×2^{16}

GATE

TCP	CN
IIT-K	IIT-KGP
IISc	IISc
IIT-B	IIT-K



Topic : TCP Timer



TCP uses five timers :

1. Retransmission Timer

2. Persistence Timer

3. Keepalive Timer

4. Time-waited Timer

5. Delayed Timer

→ For Reliable services

TCP uses it; ARQ



Topic : Retransmission Timer

Frame base Flow Control
In Data Link layer
Time-out value $\rightarrow$ static
Point-to-point Link

$\rightarrow$ Used to control a lost or discarded segment

$\rightarrow$ To Recover the loss, via ARQ.

$\rightarrow$ TCP creates a "retransmission timer" when it sends a segment

$\rightarrow$ TCP uses dynamic retransmission time,
which is different for each connection and may be changed during connection

$\Rightarrow$ Time-out value for each transmitted segment
= Updated Retransmission time



Topic : TCP Time-out Value

SampleRTT (MeasuredRTT) :

- Measured time from segment transmission until ACK receipt
- SampleRTT will vary ✓

EstimatedRTT :

Initially : EstimatedRTT = SampleRTT

$$\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$$

where $\alpha = 0.125$



Topic : TCP Time-out Value

DevRTT :

$$\text{DevRTT} = (1 - \beta) * \text{DevRTT} + \beta * |\text{SampleRTT} - \text{DevRTT}|$$

where $\beta = 0.25$

$$\text{Time-out Interval} = \text{EstimatedRTT} + 4 * \text{DevRTT}$$

[MCQ] [GATE-IT-2007, 1-Mark] *11T-K, H.W*



#Q. Consider the following statements about the timeout value used in TCP.

- i. The timeout value is set to the RTT (Round Trip Time) measured /during TCP connection establishment for the entire duration of the connection.
- ii. Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection.
- iii. Timeout value is set to twice the propagation delay from the sender to the receiver.

Which of the following choices hold?

- ☐ A (i) is false, but (ii) and (iii) are true
- ☐ B (i) and (iii) are false, but (ii) is true
- ☐ C (i) and (ii) are false, but (iii) is true
- ☐ D (i), (ii) and (iii) are false

[MCQ] [GATE-2015, Set-1, 1-Mark] JIT-K, H.W.



#Q. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are False with respect to the TCP connection?

- I. If the sequence number of a segment is m , then the sequence number of the subsequent segment is always $m+1$.
- II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.
- III. The size of the advertised window never changes during the course of the TCP connection.
- IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window

- ☐ A III only
- ☐ C I and IV only

- ☐ B I and III only
- ☐ D II and IV only



Topic : Persistence Timer



→ When TCP receives a segment with a window size of “zero”

→ TCP starts a persistence timer

→ When the persistence timer goes off,
the TCP sends a special segment called “probe”

→ This prevents from deadlock



Topic : TCP



TCP Sender

TCP Receiver

Recv Buffer

[Hold the
Transmission]*
Till Non-zero
window size
advertized

Window Size = 0
ACK

lost X
Window Size = 200
ACK

Space Available
Non-zero value
advertize.

Non-zero window
advertisement
will come

[Can lead deadlock]
This can lead
indefinite wait at
both side;

TCP Segment
will come.

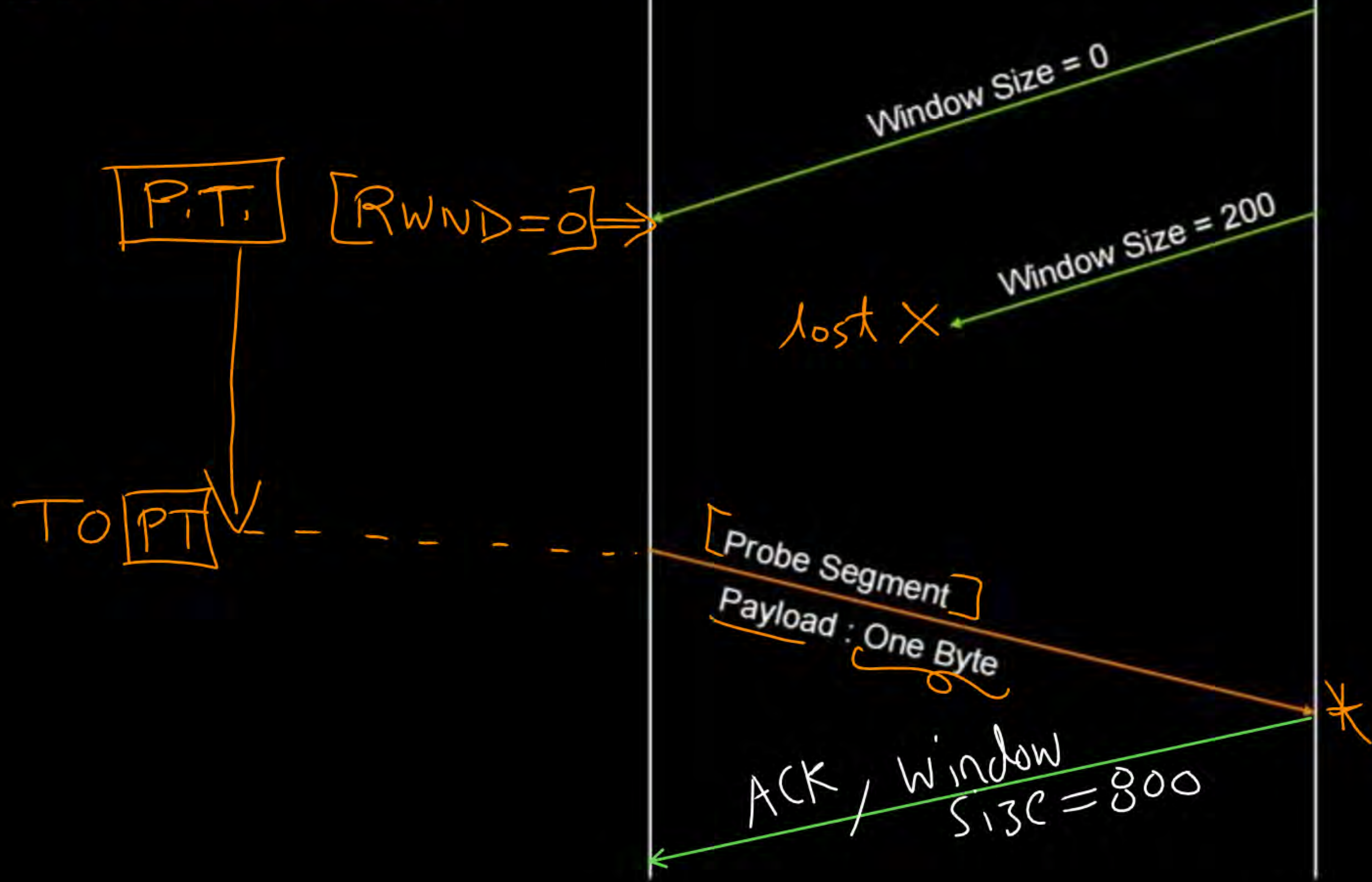


Topic : TCP



TCP Sender

TCP Receiver





2 mins Summary



Topic

TCP Options ✓

Topic

TCP Window Scale ✓

Topic

TCP Timers ⇒



THANK - YOU

